DRUGS · SPECIAL ENT AGENTS

Sclerosants

Sclerotherapy injects a substance into a vascular or lymphatic malformation to damage its lining and cause fibrosis and shrinkage. It is first-line for macrocystic lymphatic malformations of the neck (where surgery risks nerves and recurrence), a major treatment for venous malformations, and an option for ranulas. Agents differ in their indications and risks, and the procedure is usually done by interventional radiologists with ENT and multidisciplinary input.

1 CHAPTER ONE At a glance

The agents on one page.

1.1 Summary

SCLEROSANTS

	SUMMARY
OK-432 (picibanil)	Lyophilised Streptococcus pyogenes; triggers inflammation; macrocystic lymphatic malformations, ranulas; fever and swelling for days; not available everywhere
Doxycycline	10 to 20 mg/ml injected into cysts; macrocystic and mixed lymphatic malformations; pain, swelling; avoid systemic dosing concerns in small children (tooth staining is rare with intralesional use)
Bleomycin	Antitumour antibiotic; microcystic and macrocystic lymphatic malformations and venous malformations; less swelling than OK-432; cumulative dose limits for pulmonary fibrosis (small doses are used)
Sodium tetradecyl sulfate and polidocanol	Detergent sclerosants (often as foam) for venous malformations

	SUMMARY
Ethanol	Powerful sclerosant for venous and arteriovenous malformations; risk of nerve injury, skin necrosis, and cardiopulmonary collapse; usually under general anaesthesia

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Lymphatic malformations

Macrocystic lesions respond well (often more than 80% significant reduction); microcystic lesions respond poorly and may need sirolimus (an mTOR inhibitor, increasingly used for complex lymphatic malformations), surgery, or radiofrequency ablation of mucosal vesicles. Lesions of the tongue, floor of mouth and airway may swell after injection, so a planned tracheostomy or airway observation is needed for some.

2.2 Ranula

Plunging and oral ranulas recur after drainage or marsupialisation; excision of the sublingual gland is the most reliable treatment, with OK-432 sclerotherapy an alternative in some centres.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Pain, fever and marked swelling (airway risk in neck and tongue lesions), skin necrosis and ulceration, nerve injury (facial, hypoglossal, recurrent laryngeal),

infection; bleomycin pulmonary toxicity at cumulative doses; ethanol systemic toxicity.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Macrocystic versus microcystic lymphatic malformations and the choice of treatment.
- Airway risk after sclerotherapy.
- Sirolimus as a medical option.

4.2 Common pitfalls

COMMON PITFALLS

- Sclerotherapy of a tongue base malformation without an airway plan.

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